THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. $Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$ produces a precipitate of colour: 1

(a) blue **(b)** yellow **(c)** black **(d)** white

2. A student states: “Methanoic acid in nettle stings is neutralised by an acid.” The correct response is: 1

(a) Correct **(b)** It cannot be neutralised

(c) Incorrect — it is neutralised by a base **(d)** Cannot be determined

3. An aqueous solution with pH 13 is: 1

(a) strongly acidic **(b)** weakly basic **(c)** strongly basic **(d)** neutral

4. For the anode in electrolytic refining of copper, Student A says impure copper; Student B says pure copper. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

5. The green coating on copper vessels is due to: 1

(a) copper oxide **(b)** basic copper carbonate

(c) copper nitrate **(d)** copper sulphate

6. The number of carbon atoms in a molecule of propanone is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 2

7. Which compound has a triple bond? 1

(a) ethene **(b)** ethyne **(c)** methane **(d)** ethane

8. Fungi such as bread mould obtain food by: 1

(a) photosynthesis **(b)** saprotrophic nutrition

(c) holozoic nutrition **(d)** parasitic nutrition

9. Which of these is an involuntary action? 1

(a) talking **(b)** writing **(c)** heartbeat **(d)** walking

10. The method of vegetative propagation in sugarcane is: 1

(a) roots **(b)** stem cutting **(c)** seeds **(d)** leaf buds

11. The structure that protects spores of Rhizopus is: 1

(a) cilia **(b)** flagellum **(c)** thick wall **(d)** pseudopodia

12. A woman has a perfect pair of sex chromosomes called: 1

(a) XY **(b)** YY **(c)** XO **(d)** XX

13. The chromosome number is restored at: 1

(a) fertilisation **(b)** germination **(c)** growth **(d)** gamete formation

14. Which mirror is used in solar furnaces? 1

(a) concave **(b)** convex **(c)** plane **(d)** none

15. The eye lens is: 1

(a) cylindrical **(b)** convex **(c)** concave **(d)** plane

16. A student states: “Nichrome has higher resistivity than copper.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) They are equal **(d)** Cannot be determined

17. Ammeter is connected in a circuit: 1

(a) in series **(b)** anywhere **(c)** across the cell **(d)** in parallel

18. In a domestic circuit, appliances are connected: 1

(a) in series **(b)** randomly **(c)** in both **(d)** in parallel

19. Which of the following best describes a food web? 1

(a) a single food chain **(b)** energy pyramid only

(c) interconnected food chains **(d)** a trophic level

20. Which of these is a non-biodegradable pollutant? 1

(a) sewage **(b)** cow dung **(c)** paper **(d)** DDT

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Define a precipitation reaction with an example. 2

22. What is a family of salts? Give two examples. 2

23. What is the role of HCl in the stomach? 2

24. Name two tissues that respond to nerve impulses. 2

25. Why is vegetative propagation practised for growing some types of plants? 2

26. What is meant by 'pure line'? 2

27. What is meant by lateral displacement? 2

28. Why are copper wires used in solenoids rather than iron wires? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Why does copper sulphate solution change colour when an iron nail is dipped in it? What happens if a copper strip is dipped in iron sulphate solution? Why? 3

30. What is meant by the reactivity series? Arrange Na, Cu, Fe, Zn and Ag in decreasing order of reactivity. 3

31. How does the touch-me-not plant respond to touch? 3

32. Why is the reproductive system not fully developed at birth? What changes occur at puberty in general? 3

33. Why does a star appear to twinkle? Explain. 3

34. State the factors affecting the magnetic field at the centre of a circular coil. 3

35. Equal lengths of magnesium ribbon are put in test tubes A (HCl) and B (acetic acid) of the same concentration. In which tube is fizzing more vigorous and why? 3

OR

What is the difference between hydrated and anhydrous salts? Give one example of each. 3

36. What are the steps you would take to reduce the use of non-biodegradable materials in your daily life? 3

OR

Explain why some substances are biodegradable and others are not. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) An object 3 cm high is placed 24 cm from a convex lens of focal length 8 cm. Find the position, size and nature of the image. 5
(ii) State one use of a convex lens.

OR

Case study: A student used a convex lens of focal length 20 cm and moved a candle to find sharp images on a screen. 5
(i) Where should the candle be placed for a same-size image?
(ii) What happens to the image when the candle is moved to 15 cm?
(iii) Find the image distance for the candle at 30 cm.
(iv) Find magnification for (iii).
(v) Power of the lens?

38. (i) What are soaps and detergents? 5
(ii) Explain micelle formation with a diagram.
(iii) Why do detergents work in hard water?

OR

(i) Why is carbon able to form a large number of compounds? 5
(ii) Draw the electron-dot structures of methane and ethene.
(iii) Why are these compounds poor conductors?

39. (i) Derive the relation $P = I^2R = V^2/R$. 5
(ii) Two bulbs of 100 W and 60 W are connected in parallel to 220 V. Which glows brighter? Find the total current.

OR

Case study: An electric heater of 1500 W and a bulb of 100 W are used for 4 hours each at 220 V. 5
(i) Find the total energy consumed.
(ii) Find the cost at ₹7 per unit.
(iii) Find the resistance of the bulb.
(iv) Which draws more current?
(v) Why is the heater element made of nichrome?

40. (i) Explain the process of digestion of food from mouth to stomach. 5

(ii) What would happen if mucus were not secreted in the stomach?

OR

(i) Explain the role of blood as a transport medium. 5

(ii) What are the components of blood and their functions?

(iii) Why do we need a pump?

END OF QUESTION PAPER **Total: 80 Marks**

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